

Mosarruf Hossain Shawon

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Research Interests

UAS based atmospheric sensing and wind field estimation; flight dynamics and control in turbulent environments; state estimation and stochastic filtering; aircraft system identification; fixed wing UAS flight testing; wildfire induced flow sensing; wake vortex interaction; small and micro UAS development.

Education

University of Kansas, Lawrence, KS Aug. 2021 to Present
Ph.D. in Aerospace Engineering

Advisor Dr. Haiyang Chao

University of Kansas, Lawrence, KS Dec. 2025
Master of Engineering in Aerospace Engineering

Military Institute of Science and Technology, Dhaka, Bangladesh Jan. 2018
B.Sc. in Aeronautical Engineering, Major in Aerospace
Undergraduate Supervisor Dr. M. A. Taher Ali

Research Experience

Cooperative Unmanned Systems Laboratory (CUSL), University of Kansas Aug. 2021 to Present
Graduate Research Assistant

- Developed model based stochastic and EKF based estimation methods for three dimensional wind velocity, aerodynamic flow angles, and atmospheric disturbance states from fixed wing UAS flight data, with applications to wake vortex and fire plume encounters.
- Led more than 20 fixed wing UAS flight test campaigns covering experiment planning, mission design, aircraft preparation, sensor integration, on site execution, data collection, and post flight analysis.
- Investigated UAS based sensing of fire induced wind fields and the relationship between aircraft response, atmospheric disturbances, and evolving wildfire environments.
- Modified ArduPlane firmware and flight test procedures to support aircraft system identification experiments and specialized research missions.
- Extended and modified HawkWakeSim v2.1 VLM in MATLAB to simulate fixed wing UAS response during wake vortex encounters and support estimator development and validation studies.
- Validated estimator outputs against independent wind measurements from CSAT3B sonic anemometry, multi hole probe measurements, and RTK GNSS and IMU instrumentation.

Publications

Journal Articles

- Justin J. Matt, Haiyang Chao, **Mosarruf H. Shawon**, Benjamin C. Svoboda, and Steven G. Hagerott, “**System Identification for Small Flying Wing UAS Using Open Loop and Closed Loop Flight Data**,” *AIAA Journal of Aircraft*, 2025.
- **Mosarruf H. Shawon**, Matthew Rhudy, Tor Arne Johansen, Justin J. Matt, Haiyang Chao, Pengzhi Tian, Saket Gowravaram, Harold P. Flanagan, and Jacksen Goyer, “**Model Based 3D Wind Velocity Estimation during UAS Fire Plume Encounters**,” *AIAA Journal of Guidance, Control, and Dynamics*, under review.

Conferences

- **Mosarruf H. Shawon** and Haiyang Chao, “**Spatiotemporal Wind Field Sensing and Estimation using Multiple UAS in Formation,**” *AIAA Science & Technology Forum & Exposition (SciTech)*, 2027. *Accepted to appear.*
- Kelvin Tran, **Mosarruf H. Shawon**, and Haiyang Chao, “**Multi Point System Identification and Scaling for Small Flying Wing UAS,**” *AIAA Science & Technology Forum & Exposition (SciTech)*, 2027. *Accepted to appear.*
- **Mosarruf H. Shawon**, Matthew Rhudy, Justin J. Matt, Huixuan Wu, and Haiyang Chao, “**Sensing and Estimation of Fire Induced Wind Field using Small UAS,**” *AIAA Science & Technology Forum & Exposition (SciTech)*, 2026. *Selected as one of the top three Best Student Papers by the AIAA Atmospheric Flight Mechanics Technical Committee.*
- **Mosarruf H. Shawon**, Haiyang Chao, Matthew Rhudy, Tor Arne Johansen, Harold Patrick Flanagan, Pengzhi Tian, and Jacksen Goyer, “**Vertical Wind Velocity Estimation during UAS Fire Plume Encounters,**” *AIAA Science & Technology Forum & Exposition (SciTech)*, 2025.

Teaching Experience

University of Kansas, Department of Aerospace Engineering
Graduate Teaching Assistant, AE 551 Dynamics of Flight II

Spring 2022, Spring 2024, Spring 2026

- Assisted instruction in aircraft dynamic modeling and control analysis, including state space methods, stability analysis, classical flight modes, transfer function methods, and flight control system design.
- Developed custom roll and pitch axis PD controllers in ArduPlane firmware using C++, and evaluated controller performance against flight test data as instructional support for flight control design topics.
- Supported students with analytical and computational approaches to flight dynamics and control problems.

University of Kansas, Department of Aerospace Engineering
Graduate Teaching Assistant, AE 430 Aerospace Instrumentation Laboratory

Spring 2021

- Supported laboratory instruction involving pressure probes, thermocouples, strain gauges, hot wire anemometry, laser Doppler velocimetry, and flow visualization.
- Assisted with laboratory data acquisition, experimental procedures, MATLAB based analysis, and evaluation of student work.

Technical Skills

Programming	MATLAB, Python, C, C++
Estimation & Control	Extended Kalman Filtering (EKF), state estimation, aircraft system identification, flight dynamics modeling, flight control design and evaluation
Flight Testing & UAS	Fixed wing UAS flight testing, ArduPlane, Pixhawk, flight log processing, mission planning, hardware in the loop (HITL), software in the loop (SITL)
Simulation & Software	MATLAB/Simulink, LabVIEW, HawkWakeSim, flight dynamics simulation and post processing
Sensors & Hardware	RTK GNSS (Topcon Hiper V, u blox NEO M8P, F9P), IMU, multi hole probe, CSAT3B sonic anemometer, weather station, airspeed sensors, Raspberry Pi, Arduino, telemetry systems
Languages	Bengali (Native), English (Professional proficiency)

Service and Outreach

Reviewer AIAA SciTech 2027, AIAA SciTech 2026, American Control Conference 2024; *Journal of Aerospace Information Systems (JAIS)*

Teach For Bangladesh
Fellow

Nov. 2018 to Dec. 2020

Worked in a government primary school supporting efforts to reduce educational disparities and achievement gaps among students from underserved communities.